Measures of Center & Spread

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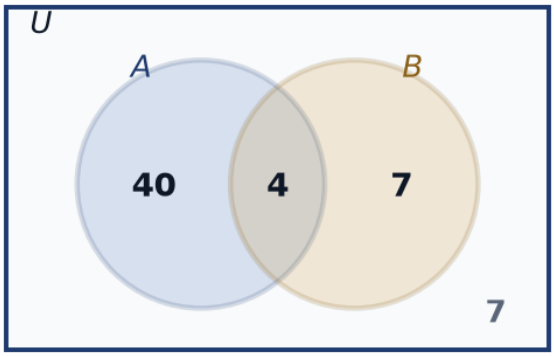
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean and median

Find the mean and median of the data set: $4, 7, 7, 9, 13$.

2 [Foundational] • Target: Mean and median

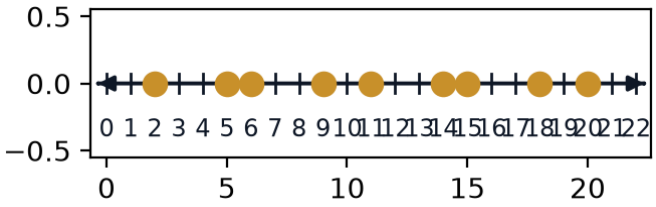
A sixth data point of 40 is added to the set $4, 7, 7, 9, 13$. Recompute the mean and median, and explain which measure changed more.



3 [Foundational] • Target: Range, IQR, and quartiles

Find the five-number summary (min, Q_1, median, Q_3, max) of the ordered data:

$2, 5, 6, 9, 11, 14, 15, 18, 20$.



4 [Foundational] • Target: Range, IQR, and quartiles

Using the five-number summary from the previous problem, find the IQR and determine whether 20 would be flagged as an outlier using the $1.5 \times$ IQR rule.

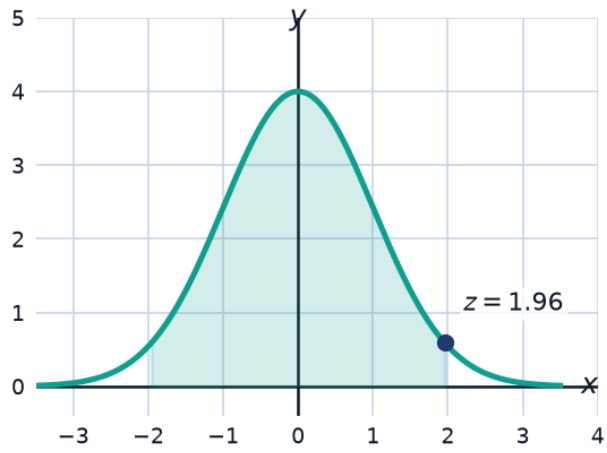
5 [Foundational] • Target: Standard deviation and variance

Find the standard deviation of the data set $2, 4, 6, 8$ (treat as a population).

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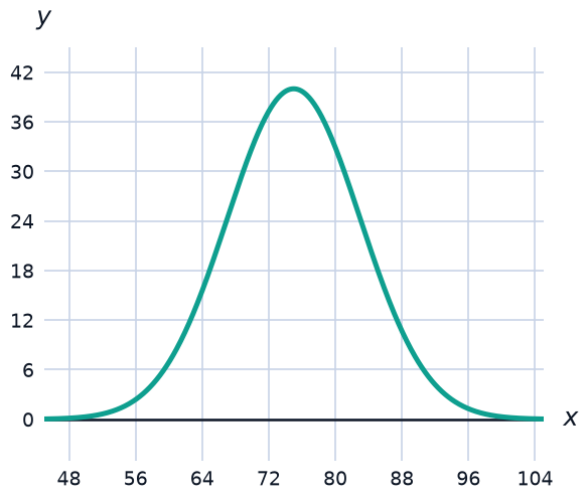
6 [Foundational] • Target: Standard deviation and variance

Explain why the standard deviation is always non-negative, and why it equals 0 only in one specific case.



7 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

The graph shows a bell-shaped distribution of exam scores with mean 75 and standard deviation 8.



a Estimate the range of scores that fall within one standard deviation of the mean.

b Estimate the range of scores that fall within two standard deviations of the mean.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing distributions using measures of spread

Class A has a mean test score of 80 with a standard deviation of 4. Class B has a mean of 80 with a standard deviation of 12. Compare the two classes' consistency, and explain what a small vs. large standard deviation indicates.

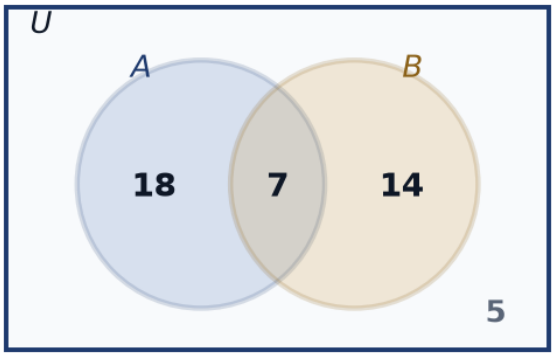
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9 [Intermediate] • Target: Resistant vs. non-resistant statistics

Explain why the median and IQR are described as "resistant" measures, while the mean and standard deviation are not, using the outlier example from earlier in this worksheet as a reference.

10 [Intermediate] • Target: Choosing appropriate summary statistics

A data set of home prices in a neighborhood is strongly skewed right due to a few very expensive mansions. Recommend which pair of summary statistics (mean/standard deviation or median/IQR) should be reported, and justify your choice.

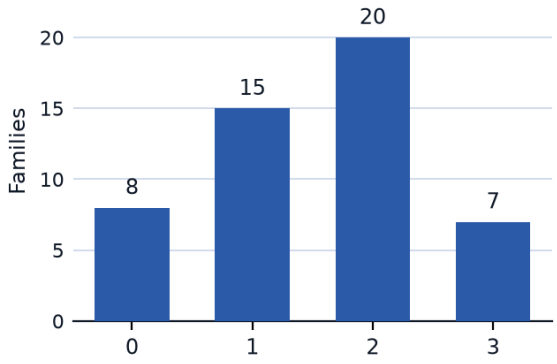


11 [Intermediate] • Target: Choosing appropriate summary statistics

A quality-control process measures the diameter of manufactured bolts, which are expected to be very consistent (low variability) and roughly symmetric. Recommend which pair of summary statistics is more appropriate here, and explain why.

12 [Intermediate] • Target: Computing summary statistics from a frequency table

A survey of families records number of children per household: 0 children (8 families), 1 child (15 families), 2 children (20 families), 3 children (7 families). Find the mean number of children per household.



13 [Intermediate] • Target: Computing summary statistics from a frequency table

A small business records weekly sales (in thousands): $10, 12, 12, 14, 17$. Find the mean, and then find the variance and standard deviation (treat as a population).

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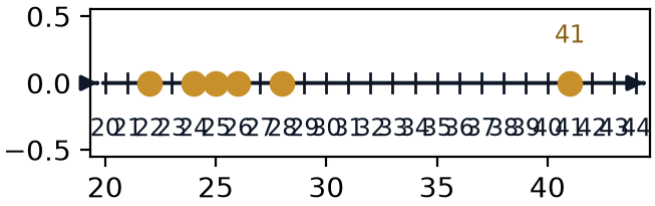
14 **[Intermediate]** • **Target: Computing summary statistics from a frequency table**

Two data sets both represent quiz scores out of 20 for two different classes of the same size, and both have the same mean of 15. Class A has a standard deviation of 1.2, while Class B has a standard deviation of 4.5. Without doing any further calculation, describe what a dotplot of each class's scores would likely look like, and explain how you can tell just from the two standard deviations.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

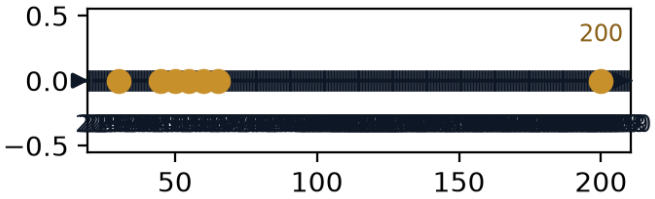
15 **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A coach records the vertical jump heights (in inches) of 7 athletes: $22, 24, 25, 26, 26, 28, 41$. Explain why the value 41 would likely be flagged as an outlier, and describe, in general terms, the two-step process for using the $1.5 \times$ IQR rule to formally check whether a value is an outlier.



16 **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

A teacher records the number of minutes 8 students spent on homework last night: $30, 45, 50, 55, 60, 60, 65, 200$. The value 200 comes from a student who mistakenly reported total weekly homework time instead of just last night. Explain, step by step, how this single incorrect data entry would affect the reported mean versus the reported median, and explain which measure a teacher should trust more when summarizing typical homework time for this data set, given the known data-entry error.



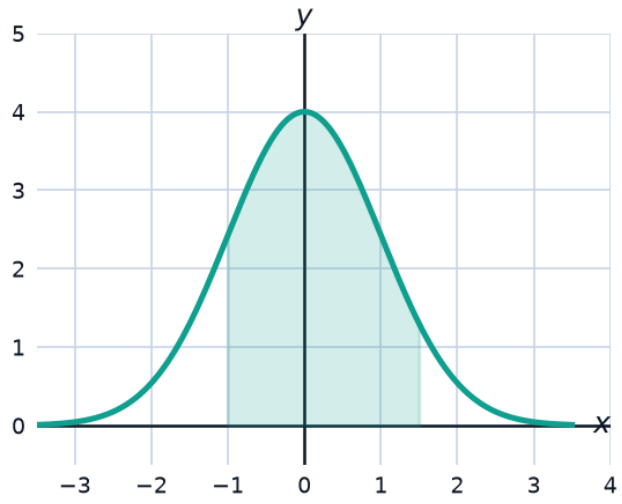
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17 **[Challenging]** • **Target: Comparing distributions using measures of spread** • **Cross-Lesson: 3.03 The Normal Distribution**

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

18 **[Challenging]** • **Target: Resistant vs. non-resistant statistics** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A set of daily high temperatures in Celsius has mean $\bar{x} = 25.0^\circ\text{C}$ and standard deviation $s = 4.0^\circ\text{C}$. Convert the temperatures to Fahrenheit using $F = 1.8C + 32$. Find the new mean and standard deviation in Fahrenheit.



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19 **[Challenging]** • **Target: Choosing appropriate summary statistics** • **Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation s_p if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

20 **[Challenging]** • **Target: Computing summary statistics from a frequency table** • **Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A marketing research firm is designing an experiment to test the effectiveness of three different advertising slogans (Slogan A, Slogan B, Slogan C) on product purchase intent. They recruit 240 participants and randomly assign them to one of the three slogan groups. After exposure to their assigned slogan, participants rate their purchase intent on a scale of 1 to 10. The experiment is designed as a completely randomized design. The data collected will be analyzed using ANOVA to compare the mean purchase intent scores across the three groups. In order to determine the minimum number of participants needed in each group for a power analysis, the researchers need to estimate the effect size. A pilot study suggests that the standard deviation of purchase intent scores within each group is approximately $\sigma = 1.8$. They aim for a minimum of $n = 80$ participants per group to achieve the desired statistical power. If the total number of participants available for this experiment were $N = 240$ and they decided to allocate the participants into $k = 3$ treatment groups, with an equal number of participants in each group, what would be the number of participants per group, n? Additionally, if the experimenters decided to use a blocked design instead, where participants are first blocked by age (Young, Middle-aged, Senior) and then randomly assigned slogans within each block, how many blocks would be needed?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Mean and median

Find the mean of the data set: $17, 28, 12, 6, 13$.

a. $\frac{81}{5}$

b. $\frac{71}{5}$

c. 19

d. $\frac{76}{5}$

22 [Foundational] • Target: Range, IQR, and quartiles

Find the mean of the data set: $1, 8, 15, 8, 7$.

a. $\frac{39}{5}$

b. $\frac{44}{5}$

c. $\frac{39}{4}$

d. $\frac{34}{5}$

23 [Foundational] • Target: Standard deviation and variance

Find the mean of the data set: $15, 26, 21, 10, 14$.

a. $\frac{43}{2}$

b. $\frac{91}{5}$

c. $\frac{86}{5}$

d. $\frac{81}{5}$

24 [Foundational] • Target: Visualizing spread with a normal-shaped distribution

Find the mean of the data set: $15, 23, 24, 25, 9$.

a. $\frac{101}{5}$

b. $\frac{91}{5}$

c. $\frac{96}{5}$

d. 24

25 [Foundational] • Target: Comparing distributions using measures of spread

Find the mean of the data set: $22, 24, 23, 15, 27$.

a. $\frac{106}{5}$

b. $\frac{111}{4}$

c. $\frac{111}{5}$

d. $\frac{116}{5}$

26 [Foundational] • Target: Resistant vs. non-resistant statistics

Find the mean of the data set: $26, 7, 1, 1, 20$.

a. 11

b. 12

c. $\frac{55}{4}$

d. 10

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27 [Foundational] • Target: Choosing appropriate summary statistics

Find the mean of the data set: $12, 15, 25, 7, 25$.

a. $\frac{89}{5}$

b. 21

c. $\frac{84}{5}$

d. $\frac{79}{5}$

28 [Foundational] • Target: Computing summary statistics from a frequency table

Find the median of the data set: $16, 15, 5, 25, 29, 12$.

a. $\frac{31}{2}$

b. 17

c. $\frac{33}{2}$

d. $\frac{37}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mean, Median, and Skewness

Find the median of the data set: $31, 29, 29, 23, 23, 36$.

a. 30

b. 27

c. $\frac{57}{2}$

d. 29

30 [Intermediate] • Target: The 1.5 x IQR Rule for Outliers

Find the median of the data set: $12, 30, 16, 1, 22, 37$.

a. 21

b. 19

c. $\frac{59}{3}$

d. 20

31 [Intermediate] • Target: Standard Deviation and Variance

Find the median of the data set: $24, 30, 40, 6, 31, 5$.

a. $\frac{37}{2}$

b. 27

c. $\frac{68}{3}$

d. 28

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32 [Intermediate] • Target: Comparing Distributions with Boxplots

Find the median of the data set: $25, 32, 13, 21, 40, 31$.

a. $\frac{53}{2}$

b. 29

c. 27

d. 28

33 [Intermediate] • Target: Mean and median

Find the median of the data set: $10, 13, 40, 18, 9, 14$.

a. $\frac{52}{3}$

b. 14

c. $\frac{27}{2}$

d. $\frac{29}{2}$

34 [Intermediate] • Target: Range, IQR, and quartiles

Find the median of the data set: $22, 12, 30, 18, 11, 28$.

a. $\frac{121}{6}$

b. $\frac{39}{2}$

c. 21

d. 20

35 [Intermediate] • Target: Standard deviation and variance

Find the IQR of the ordered data set: $2, 8, 10, 12, 20, 23, 34, 45$.

a. $\frac{39}{2}$

b. $\frac{41}{2}$

c. $\frac{37}{2}$

d. $\frac{57}{2}$

36 [Intermediate] • Target: Visualizing spread with a normal-shaped distribution

Find the IQR of the ordered data set: $18, 21, 35, 38, 39, 46, 55, 58$.

a. $\frac{43}{2}$

b. $\frac{47}{2}$

c. $\frac{45}{2}$

d. $\frac{101}{2}$

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37 [Intermediate] • Target: Comparing distributions using measures of spread

A data set has $Q_1 = 24$ and $Q_3 = 46$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

a. 78

b. 68

c. 79

d. 80

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Resistant vs. non-resistant statistics • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A data set has $Q_1 = 26$ and $Q_3 = 48$. Using the $1.5 \times$ IQR rule, find the upper fence used to detect outliers.

a. 80

b. 70

c. 81

d. 82

39 [Challenging] • Target: Choosing appropriate summary statistics • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

Find the standard deviation of the data set $29, 22, 31, 38$ (treat as a population), to two decimal places.

a. 6.30

b. 5.40

c. 5.70

d. 6.00

40 [Challenging] • Target: Computing summary statistics from a frequency table • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Find the standard deviation of the data set $22, 20, 16, 18$ (treat as a population), to two decimal places.

a. 2.54

b. 2.24

c. 2.84

d. 1.94

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41 [Challenging] • Target: Mean, Median, and Skewness • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

In a strongly right-skewed distribution of household incomes, which relationship between the mean and median is typically expected?

a. The mean is greater than the median

b. The mean is less than the median

c. The mean equals the median

d. The relationship cannot be determined without sample size

42 [Challenging] • Target: Mean, Median, and Skewness • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A dataset has values: $3, 5, 7, 8, 9, 11, 15$. If the maximum value 15 is replaced with 50, which statistic will NOT change?

a. Median

b. Mean

c. Range

d. Standard Deviation

43 [Challenging] • Target: The 1.5 x IQR Rule for Outliers • Cross-Lesson: 2.01 Sampling Methods & Bias

What percentage of observations in any distribution lies strictly between Q_1 and Q_3?

a. Approximately 50%

b. 25%

c. 75%

d. 68%

44 [Challenging] • Target: Standard Deviation and Variance • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

If every observation in a sample has the exact same value of 14, what is the sample standard deviation?

a. 0

b. 14

c. 1

d. Undefined

45 [Challenging] • Target: Comparing Distributions with Boxplots • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A boxplot shows a long right whisker, a right-half box wider than the left-half box, and no outliers. What is the shape of the distribution?

a. Skewed to the right

b. Skewed to the left

c. Roughly symmetric

d. Uniform

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